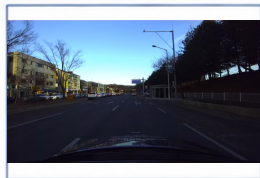
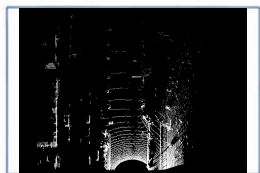


(a) Existing fusion

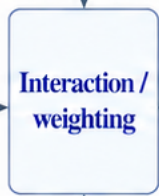
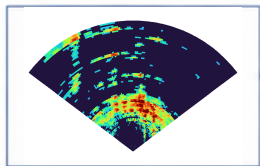
Camera



LiDAR



4D radar



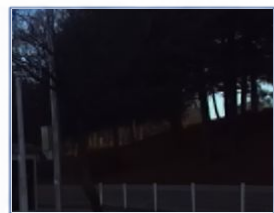
Fused features



Shared / specific roles remain implicit

(b) Why decouple?

Background can agree



Camera



LiDAR



4D radar

Object cues can differ



Camera



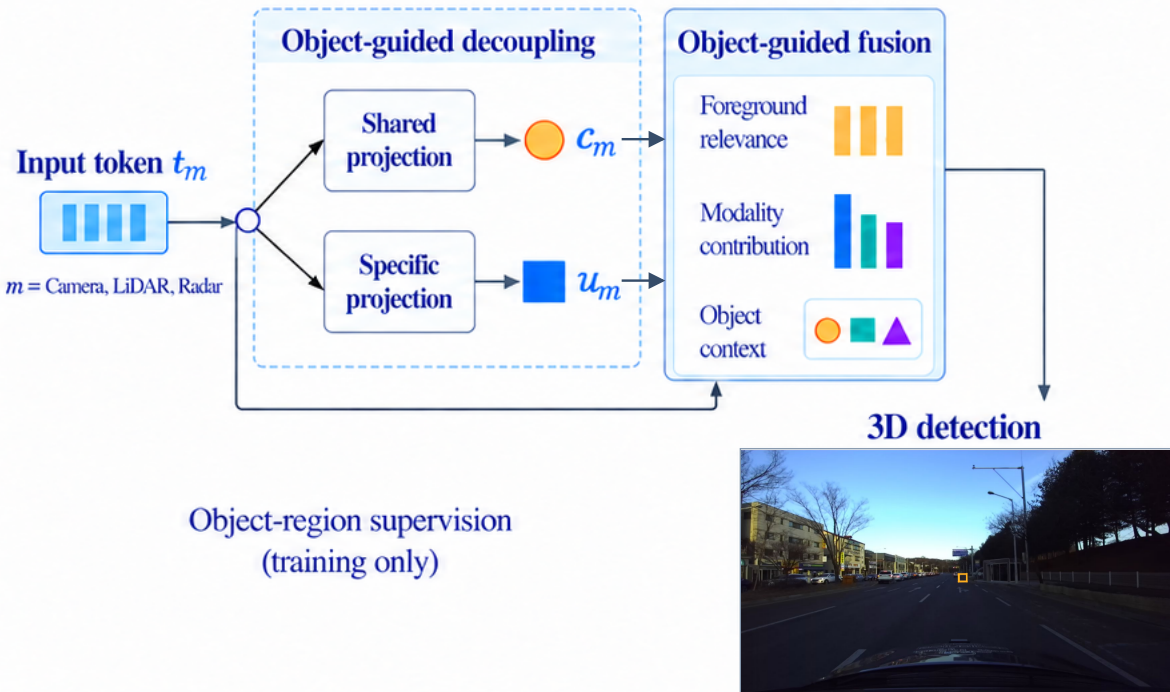
LiDAR



4D radar

Agreement alone is insufficient

(c) ObjDec (ours)



Explicit representation roles guide fusion